

Roll No.:**Set I****B. Tech./CA-1/2024-25****Time: 45 Mints****Digital Signal Processing (ECC- 402)****Full Marks: 3*5=15****Question 1:** Find the FT of the following:

I. $\left(\frac{1}{2}\right)^{n-1} u(n-1)$

Question 2: Determine the convolution of the pairs of signals by means of the z-transform $x_1(n) =$

$$\left(\frac{1}{2}\right)^n u(n); x_2(n) = \cos(\pi n) u(n)$$

Question 3: For each impulse response listed below, determine whether the corresponding system is (i) causal (ii) stable.

a. $h(n) = 2^n u(-n)$

Roll No.:**Set II****B. Tech./CA-1/2024-25****Time: 45 Mints****Digital Signal Processing (ECC- 402)****Full Marks: 3*5=15****Question 1:** Find the FT of the following:

I. $\left(\frac{1}{2}\right)^{|n-1|}$

Question 2: Evaluate the step response for the LTI System represented by the following impulse response.

$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$

Question 3: The step response of an LTI System is $s(n) = \left(\frac{1}{3}\right)^{n-1} u(n+2)$; Find the $H(z)$ and $h(n)$.**Roll No.:****Set III****B. Tech./CA-1/2024-25****Time: 45 Mints****Digital Signal Processing (ECC- 402)****Full Marks: 3*5=15****Question 1:** Two causal systems with impulse responses $h_1(n) = a\delta(n) + \delta(n-1)$ and $h_2(n) = b^n u(n)$; $|b| < 1$ are connected in cascade. Determine the frequency response $H(e^{j\omega})$ of the overall system. Find the values of “a” and “b” for $|H(e^{j\omega})| = 1$.**Question 2:** For each impulse response listed below, determine whether the corresponding system is (i) causal (ii) stable.

$$h(n) = \delta(n) + \sin(\pi n)$$

Question 3: Determine the inverse Z-transform by the partial fraction expansion method.

$$X(z) = \frac{z+2}{2z^2-7z+3}$$